Constructive and Destructive Training Regimes

1 Introduction

Training large language models (LLMs) relies on adaptive optimization, momentum, gradient clipping, learning-rate schedules and other heuristics to stabilize learning and mitigate undesired behavior such as loss spikes and oscillatory updates. Yet, despite their effectiveness, it is not fully understood why these mechanisms work, and under what conditions they fail.

To address this gap, we characterize a measurable, persistent behaviour of transition between two distinct learning regimes. Early in training, successive gradients tend to align, allowing updates to accumulate constructively as the loss decreases. Later, gradient directions reverse more frequently, causing successive updates to partially cancel one another. Despite this increased cancellation, the loss can continue to decrease.

Two complementary intuitions help explain this behavior. From the perspective of representation learning, each new mini-batch updates parameters that encode previously learned structure. Depending on its direction, an update may reinforce existing representations or partially undo them. From optimization geometry, a step may overshoot the bottom of a loss valley, prompting subsequent steps to reverse direction and partially retrace the path. Both mechanisms can give rise to oscillatory parameter updates even as optimization continues to make progress.

We formalize these dynamics using identities for stochastic gradient descent (SGD), connecting the alignment of consecutive gradients to curvature and gradient noise, and cumulative update cancellation to net parameter displacement. Building on these relations, we introduce scalable diagnostics to quantify and detect these dynamics and track the transition between constructive and oscillatory regimes during large-scale model training.

2 Parameter displacement under stochastic gradient descent

Let $\theta_t \in \mathbb{R}^p$ denote the parameters after update t, and let g_t denote a mini-batch gradient evaluated at θ_{t-1}. We consider a plain SGD with learning rate $\eta > 0$,

$$\theta_t = \theta_{t-1} - \eta g_t. \tag{1}$$

After T updates, summing Eq. (1) gives the displacement from initialization,

$$\Delta_T := \theta_T - \theta_0 = -\eta \sum_{t=1}^{T} g_t. \tag{2}$$

Expanding its squared norm separates individual gradient magnitudes from their temporal interactions

$$\|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 + 2\eta^2 \sum_{1 \le u < t \le T} g_u^\top g_t. \tag{3}$$

The gradient-energy terms $\|g_t\|^2$ are nonnegative. Each pairwise inner product $g_u^\top g_t$ can be positive, zero, or negative: aligned gradients reinforce their contribution to the sum, whereas opposing gradients cancel part of it. To practically measure the alignement of nonzero consecutive gradients, define their cosine similarity as

$$\rho_t := \frac{g_{t-1}^\top g_t}{\|g_{t-1}\| \, \|g_t\|}, \qquad t = 2, \dots, T. \tag{4}$$

Separating adjacent pairs from the remaining interactions yields

$$\begin{aligned} \|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 \\ + 2\eta^2 \underbrace{\sum_{t=2}^{T} \|g_{t-1}\| \, \|g_t\| \, \rho_t}_{\text{lag-one interaction}} \\ + 2\eta^2 \underbrace{\sum_{t=3}^{T} \sum_{u=1}^{t-2} g_u^\top g_t}_{\text{interaction beyond lag one}} . \end{aligned} \tag{5}$$

Negative ρ_t contributes cancellation between consecutive updates. However, longer-lag interactions can reinforce or offset these negative contributions. Thus, adjacent anti-alignment diagnoses a local interaction, while the complete double sum determines longer-term cumulative interaction.

2.1 Curvature, gradient noise, and consecutive alignment

For mini-batch SGD, we explicitly incorporate stochastic gradient noise and examine its contribution to their directional alignment. Let $\mathcal{L}$ denote the full training objective. We write a stochastic gradient as

$$g_t = \nabla \mathcal{L}(\theta_{t-1}) + \xi_t, \tag{6}$$

where ξ_t is the gradient estimation error. Suppose $\mathcal{L}$ is twice continuously differentiable along the step under consideration, and define $H_{t-1} := \nabla^2 \mathcal{L}(\theta_{t-2})$. Expanding the full gradient around θ_{t-2} and substituting the preceding SGD step gives

$$g_t = (I - \eta H_{t-1}) g_{t-1} + (\xi_t - \xi_{t-1}) + r_t, \tag{7}$$

where r_t is the Taylor remainder. Neglecting r_t recovers the first-order approximation. Taking the inner product with g_{t-1} then gives

$$\rho_t \approx \frac{\underbrace{\|g_{t-1}\|^2}_{\text{Gradient magnitude}} - \underbrace{\eta g_{t-1}^\top H_{t-1} g_{t-1}}_{\text{Curvature contribution}} + \underbrace{g_{t-1}^\top (\xi_t - \xi_{t-1})}_{\text{Noise contribution}}}{\|g_{t-1}\| \|g_t\|}. \tag{8}$$

The numerator contains the preceding gradient energy, the curvature, and the gradient noise contributions. Positive curvature along the preceding gradient direction reduces its alignment with the next gradient. The noise difference can reinforce or oppose this effect on an individual step. We normalize the curvature and noise terms by the preceding gradient energy

$$\kappa_t := \frac{g_{t-1}^\top H_{t-1} g_{t-1}}{\|g_{t-1}\|^2}, \qquad \varepsilon_t := \frac{g_{t-1}^\top (\xi_t - \xi_{t-1})}{\|g_{t-1}\|^2}. \tag{9}$$

With these definitions, Eq. (8) becomes

$$\boxed{\rho_t \approx \frac{\|g_{t-1}\|}{\|g_t\|}(1 - \eta\kappa_t + \varepsilon_t).} \tag{10}$$

The norm ratio is positive, so the first-order sign prediction depends on $1 - \eta\kappa_t + \varepsilon_t$ (Table 1).

Table 1: Consecutive-gradient alignment predicted by the first-order model.

Alignment	Cosine sign	First-order condition
Positive	$\rho_t > 0$	$\eta\kappa_t < 1 + \varepsilon_t$
Orthogonal	$\rho_t = 0$	$\eta\kappa_t = 1 + \varepsilon_t$
Negative	$\rho_t < 0$	$\eta\kappa_t > 1 + \varepsilon_t$

2.1.1 Directional curvature

The quantity κ_t is a Hessian Rayleigh quotient along the preceding stochastic gradient. If $H_{t-1}v_i = \lambda_i v_i$ in an orthonormal eigenbasis and $g_{t-1} = \sum_i a_i v_i$, then

$$\kappa_t = \frac{\sum_i \lambda_i a_i^2}{\sum_i a_i^2}. \tag{11}$$

Thus, the gradient selects an eigenvalue-weighted average rather than the maximum eigenvalue. A sharp direction can exist without contributing much to the current gradient. Rearranging Eq. (10) motivates the observable quantity

$$\eta\widehat{\kappa}_t := 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|} + \varepsilon_t. \tag{12}$$

For mini-batch gradients, the noise term requires additional information and cannot be recovered from adjacent cosine and norms alone. For full-batch GD, $\xi_t = 0$ and hence $\varepsilon_t = 0$. The recorded gradient-only quantity is then exactly

$$\eta\widehat{\kappa}_t = 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|} = -\frac{g_{t-1}^\top (g_t - g_{t-1})}{\|g_{t-1}\|^2}. \tag{13}$$

It measures the change in the gradient projected onto its preceding direction.

2.2 Normalized displacement and cumulative interference

Adjacent alignment concerns one pair of steps. To characterize accumulation over the complete trajectory, we compare the sum of gradient lengths with the length of their sum

$$A_T := \sum_{t=1}^{T} \|g_t\|, \qquad B_T := \left\| \sum_{t=1}^{T} g_t \right\|. \tag{14}$$

For Eq. (1), ηA_T is the discrete path length and $\eta B_T = \|\Delta_T\|$ is its endpoint displacement. Provided $A_T > 0$, we define

$$\boxed{R_T := \frac{B_T}{A_T} = \frac{\|\Delta_T\|}{\eta \sum_{t=1}^{T} \|g_t\|} \in [0, 1].} \tag{15}$$

The numerator B_T is the net displacement; the denominator A_T is the total length of all SGD steps. This normalized displacement R_T expresses how much of the accumulated step length remains as net motion of parameters after the gradient directions reinforce or cancel one another.

2.3 Full-batch GD evidence on CIFAR-10

We examine the small CIFAR-10 squared-loss setting used in the released edge-of-stability experiments [1, 2]. We use the first 5,000 CIFAR-10 training images and a fully connected 3072–200–200–10 network with tanh hidden activations and biases (656,810 parameters). Four GD runs share the same frozen initialization from seed zero and use learning rates $\eta \in \{0.004, 2/300, 0.01, 0.02\}$. Each run completes 20,000 updates with no momentum, weight decay, clipping, augmentation, learning-rate schedule, or injected gradient noise. We record loss and raw-gradient diagnostics at every update. The largest Hessian eigenvalue is estimated on all 5,000 examples at initialization and every 100 updates. Figure 1 shows all recorded values.

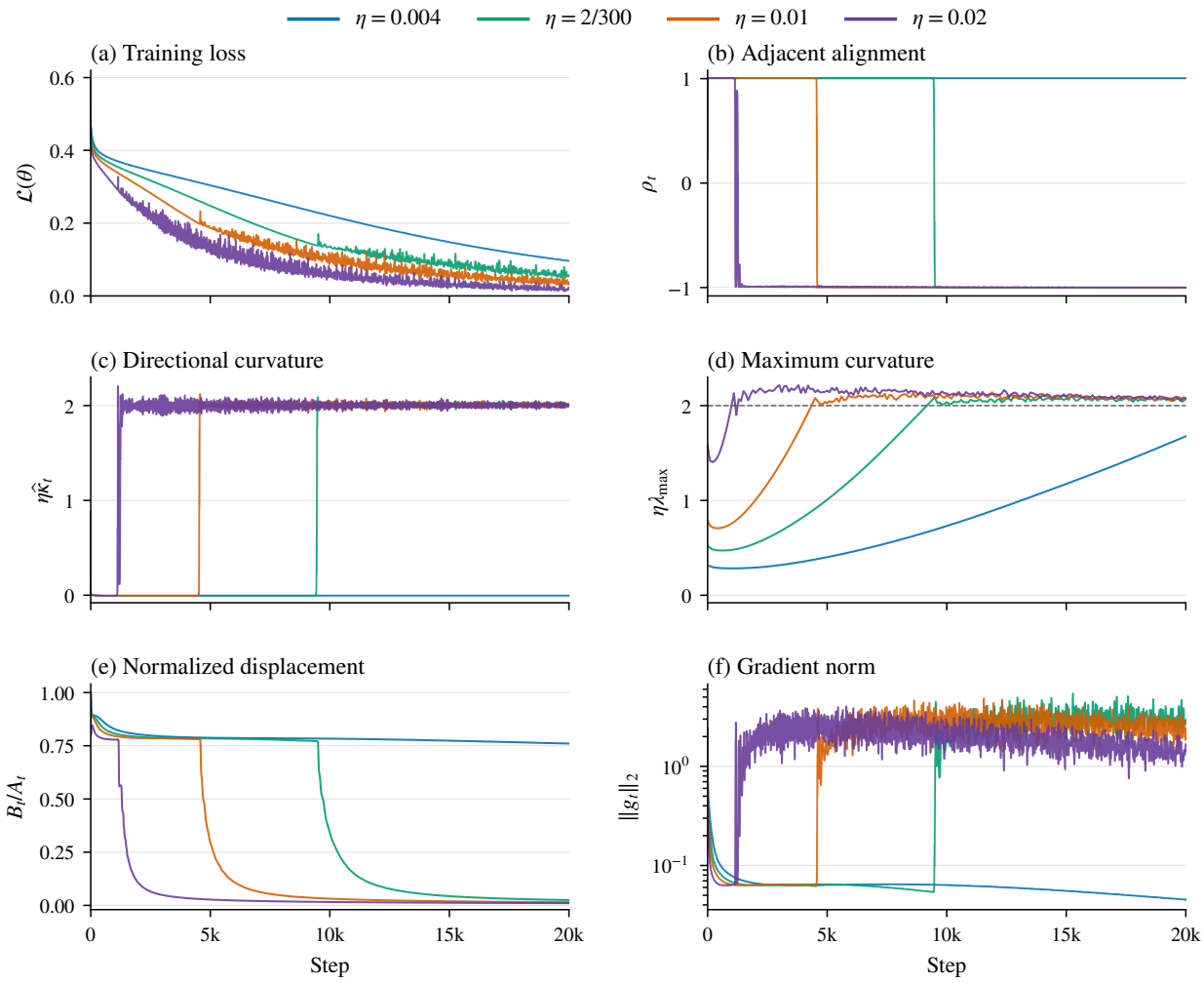


Figure 1: Full-batch GD on CIFAR-10 with four matched learning rates. (a) Training loss; (b) consecutive-gradient cosine; (c) the gradient-derived quantity $\eta\widehat{\kappa}_t = 1-\rho_t \|g_t\| / \|g_{t-1}\|$; (d) the scaled largest algebraic Hessian eigenvalue, with the dashed reference at two; (e) full-prefix normalized displacement B_t/A_t; and (f) raw-gradient norm on log Y. All curves are unsmoothed. Hessians are measured every 100 updates; the other diagnostics are measured every update. Larger learning rates enter the alternating-gradient regime earlier. Loss decreases over the horizon while cumulative cancellation and oscillatory gradient magnitudes increase.

Figure 1 shows that larger learning rates lead to an earlier transition from constructive to destructive interference between consecutive updates (panel b). This transition is accompanied by a sharp rise and peak in the estimated directional curvature $\eta\widehat{\kappa}_t$ (panel c), with $\eta\lambda_{\max}$ (panel d) near the critical GD scale of two. The correspondence between these curvature diagnostics links the observed transition to the high-curvature spectral regime in these runs. The accompanying abrupt drop of B_t/A_t toward zero (panel e) shows that most of the accumulated path length is cancelled, leaving only a small fraction as net parameter displacement. Finally, gradient norms grow and fluctuate (panel f), a behavior that our current formulation of alignment and displacement does not explain.

TODO: an alignment-guided learning-rate intervention. Test a schedule that halves the learning rate when consecutive-gradient cosine becomes negative, specifying the trigger and repeat rule before execution. Compare loss reduction at a matched update budget with fixed-rate controls. The intervention may restore positive alignment without improving convergence; these two outcomes must be evaluated separately.

2.4 Noise contribution under mini-batch sampling

Under fresh, conditionally unbiased mini-batch sampling, we have

$$\mathbb{E}[g_{t-1}^{\top}\xi_t] = 0. \tag{16}$$

But because g_{t-1} already contains the previous noise ξ_{t-1},

$$\mathbb{E}[g_{t-1}^{\top}\xi_{t-1}] = \mathbb{E}[\|\xi_{t-1}\|^2]. \tag{17}$$

Subtracting these two expectations yields

$$\mathbb{E}[g_{t-1}^{\top}(\xi_t - \xi_{t-1})] = -\mathbb{E}[\|\xi_{t-1}\|^2] \le 0. \tag{18}$$

Even unbiased noise therefore produces a nonpositive expected contribution to the unnormalized adjacent inner product. It suggests a systematic mechanism through which stochastic gradient noise can reduce successive-gradient alignment, independently of the local directional curvature.

To isolate this sampling effect, fix the parameters and write $F = N^{-1}\sum_i \ell_i$, with unit gradients $h_i = \nabla\ell_i(\theta)$ and complete objective gradient $\mu = \nabla F(\theta)$. For a uniform batch of b objective units sampled without replacement, the gradient-noise variance is

$$V_b := \mathbb{E}\left\|g^{(b)} - \mu\right\|^2 = \left(\frac{1}{b} - \frac{1}{N}\right)\frac{\sum_{i=1}^{N}\|h_i - \mu\|^2}{N-1}. \tag{19}$$

Two independently sampled gradients at these fixed parameters satisfy

$$\mathbb{E}[(g^{(b)})^{\top}g'^{(b)}] = \|\mu\|^2, \qquad \mathbb{E}\left\|g^{(b)}\right\|^2 = \|\mu\|^2 + V_b. \tag{20}$$

Noise increases gradient energy without increasing the expected consecutive inner product. The expected cosine also depends on its random denominators; it is not the ratio of the expectations in Eq. (20). For independent draws at fixed parameters with $0 < \mathbb{E}\left\|g^{(b)}\right\| < \infty$, the gradient accumulation ratio obeys

$$\frac{B_T}{A_T} \xrightarrow[T\to\infty]{\text{a.s.}} \frac{\|\mu\|}{\mathbb{E}\left\|g^{(b)}\right\|}, \tag{21}$$

Conditional Jensen's inequality for nested uniform batches shows that larger batches reduce $\mathbb{E}\left\|g^{(b)}\right\|$. Smaller batches therefore reduce this long-sequence accumulation limit at fixed parameters. During SGD, batch size also changes the trajectory and signal gradient, so the same ordering need not hold.

We isolate this sampling effect in three EOS settings: a tanh MLP on 5,000 CIFAR-10 images, ResNet-32 on all 50,000 CIFAR-10 images, and the small word-level Transformer on the complete prepared WikiText-2 corpus. For each of three initializations, every batch condition uses 32 independent sequences of 128 gradient draws. We measure noise directly as $\xi = g^{(b)} - \mu$, using the complete objective gradient as the reference. ResNet samples whole fixed groups of 1,000 images to preserve its ghost-BatchNorm objective. No parameter updates are applied: B/A describes sampled gradient accumulation in this control, rather than actual parameter displacement.

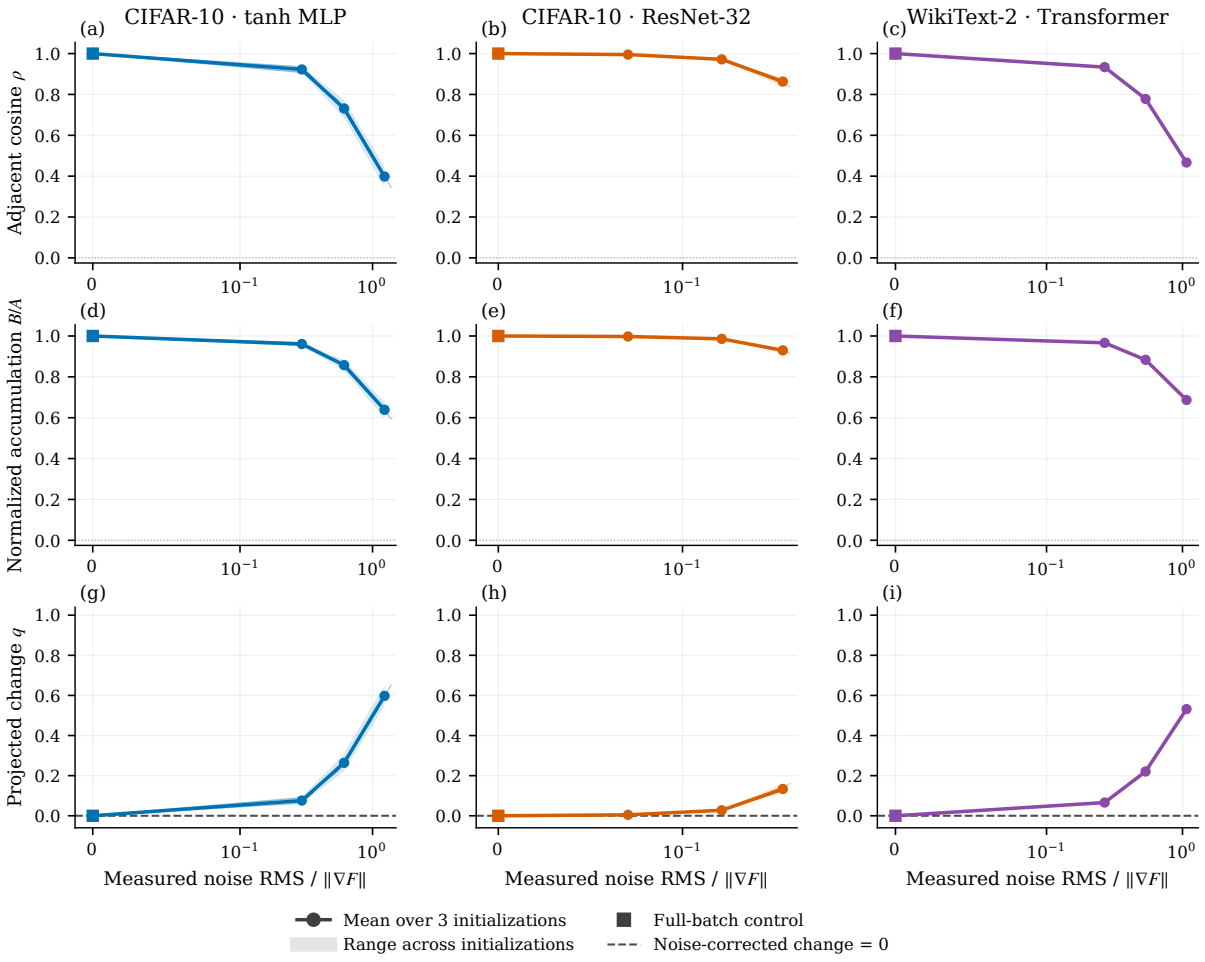


Figure 2: Real-data gradient-noise control at frozen initial parameters. Columns show CIFAR-10/MLP, CIFAR-10/ResNet-32 and WikiText-2/Transformer. Rows show mean adjacent cosine, accumulation B/A over 128 draws, and gradient-only projected change $q = 1 - g_{t-1}^{\top} g_t / \|g_{t-1}\|^2$. The horizontal axis measures noise RMS relative to the complete-gradient norm. Circles average three initializations; shading spans their range, rather than a confidence interval. Squares mark the full-batch control. The noise-corrected projected change is zero up to roundoff because the parameters and signal gradient are fixed (dashed line). This separates sampling-induced gradient changes from changes due to parameter motion.

3 Extending the SGD formulation to momentum

For Adam with constant learning rate $\eta > 0$ and no weight decay,

$$\theta_t = \theta_{t-1} - \eta d_t, \qquad d_t := \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon}, \tag{22}$$

where the operations in d_t are elementwise and

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, \quad \widehat{m}_t = \frac{m_t}{1-\beta_1^t}, \\ v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t^{\odot 2}, \quad \widehat{v}_t = \frac{v_t}{1-\beta_2^t}. \end{aligned} \tag{23}$$

Here $m_0 = v_0 = 0$, $\beta_1, \beta_2 \in [0, 1)$, and $\epsilon > 0$. The displacement expansion becomes

$$\|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|d_t\|^2 + 2\eta^2 \sum_{1 \le u < t \le T} d_u^\top d_t. \tag{24}$$

Although d_t determines parameter displacement, we study the stored first moment m_t to separate temporal gradient coherence from adaptive preconditioning. Gradients still evolve along the actual Adam trajectory.

3.1 Consecutive-momentum alignment

For nonzero momentum norms and $t \ge 2$, the recurrence gives the exact identity

$$c_t := \frac{m_{t-1}^\top m_t}{\|m_{t-1}\| \, \|m_t\|} = \frac{\beta_1 \|m_{t-1}\|^2 + (1-\beta_1) m_{t-1}^\top g_t}{\|m_{t-1}\| \, \|m_t\|}. \tag{25}$$

Using the earlier gradient-noise notation, a first-order expansion along the Adam step, neglecting the Taylor remainder, yields

$$g_t \approx g_{t-1} - \eta H_{t-1} d_{t-1} + (\xi_t - \xi_{t-1}). \tag{26}$$

Define the momentum-tracking, curvature and noise contributions as

$$\begin{aligned} a_t &:= \frac{m_{t-1}^\top (g_{t-1} - m_{t-1})}{\|m_{t-1}\|^2}, \\ \kappa_t^{(m)} &:= \frac{m_{t-1}^\top H_{t-1} d_{t-1}}{\|m_{t-1}\|^2}, \\ \varepsilon_t^{(m)} &:= \frac{m_{t-1}^\top (\xi_t - \xi_{t-1})}{\|m_{t-1}\|^2}. \end{aligned} \tag{27}$$

Substitution into Eq. (25) gives

$$\boxed{c_t \approx \frac{\|m_{t-1}\|}{\|m_t\|} \left[1 + (1-\beta_1) \left(a_t - \eta \kappa_t^{(m)} + \varepsilon_t^{(m)} \right) \right].} \tag{28}$$

The factor $1 - \beta_1$ scales these contributions to alignment. Unlike SGD, $\kappa_t^{(m)}$ is generally not a Hessian Rayleigh quotient: m_{t-1} and the preconditioned direction d_{t-1} need not be parallel.

3.2 Normalized momentum accumulation

With $A_T^{(m)} := \sum_{t=1}^{T} \|m_t\| > 0$ and $B_T^{(m)} := \left\|\sum_{t=1}^{T} m_t\right\|$, define

$$\boxed{R_T^{(m)} := \frac{B_T^{(m)}}{A_T^{(m)}} = \frac{\left\|\sum_{t=1}^{T} m_t\right\|}{\sum_{t=1}^{T} \|m_t\|} \in [0,1].} \tag{29}$$

Values near one indicate reinforcement; smaller values indicate cumulative cancellation. This describes momentum accumulation, while Adam displacement depends on d_t. Bias correction leaves c_t unchanged but generally changes the cumulative ratio, so accumulation of m_t and $\widehat{m}_t$ must be distinguished.

4 Controlled Rosenbrock experiments

We compare three deterministic AdamW trajectories on $f(x, y) = (1-x)^2 + 100(y-x^2)^2$, whose minimum is $(1, 1)$. All runs start at $(0, 1)$ and perform 2,000 float64 updates at $\eta \in \{10^{-2}, 10^{-2.5}, 10^{-3}\}$, with $(\beta_1, \beta_2) = (0.9, 0.95)$, $\epsilon = 10^{-8}$, weight decay 0.1, and exact gradients. With zero gradient noise, rearranging Eq. (28) gives the momentum estimate

$$\begin{aligned} \eta \widehat{\kappa}_t^{(m)} &:= a_t + \frac{1 - c_t \left\| m_t \right\| / \left\| m_{t-1} \right\|}{1 - \beta_1} \\ &= \frac{m_{t-1}^\top (g_{t-1} - g_t)}{\left\| m_{t-1} \right\|^2}. \end{aligned} \tag{30}$$

Both gradient-derived estimates follow the AdamW trajectory; adaptive updates, decay and finite-step effects prevent identifying them with $\lambda_{\max}$. All paths enter the curved valley and approach the minimum (Fig. 3). Larger learning rates reach raw-gradient reversal earlier, whereas momentum remains positively aligned on average (Fig. 4). Cumulative B/A stays high: late reversal therefore need not imply substantial cancellation over the full trajectory. The smallest learning rate attains the lowest final objective.

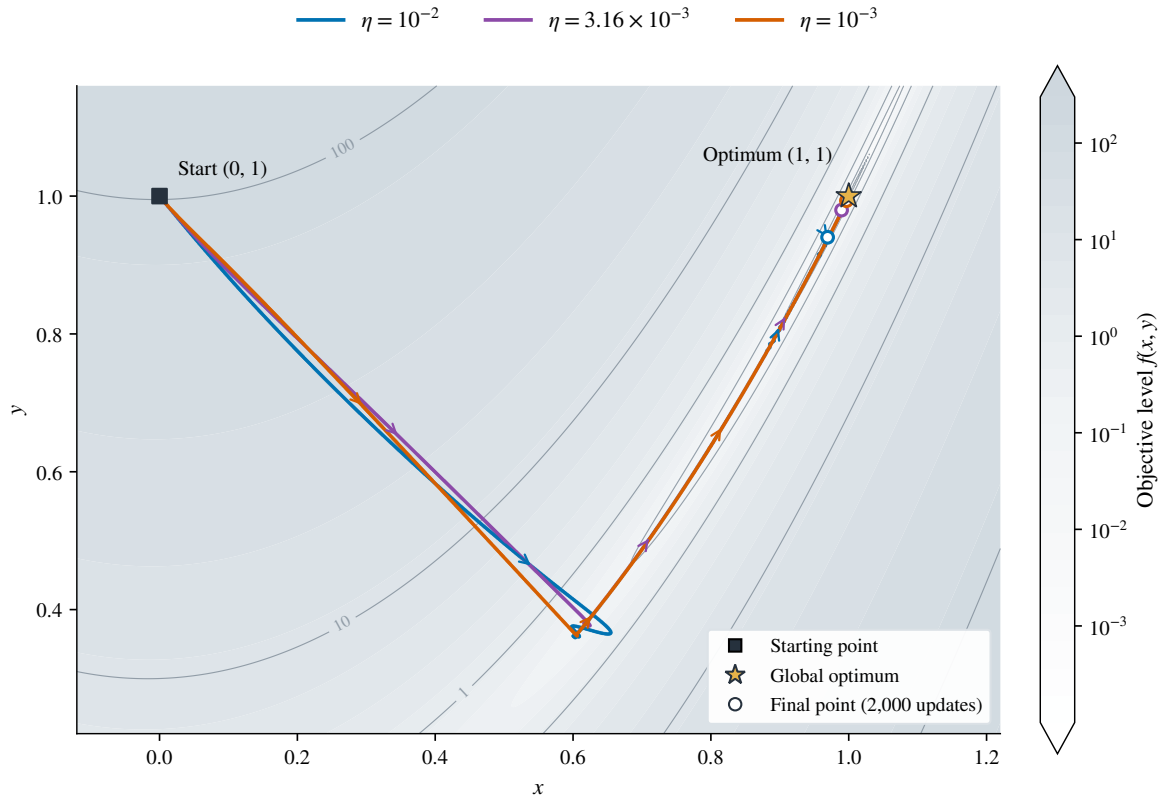


Figure 3: Complete AdamW trajectories on Rosenbrock objective level curves. The square and star mark the common start and global minimum; circles mark final states. Arrows follow update order.

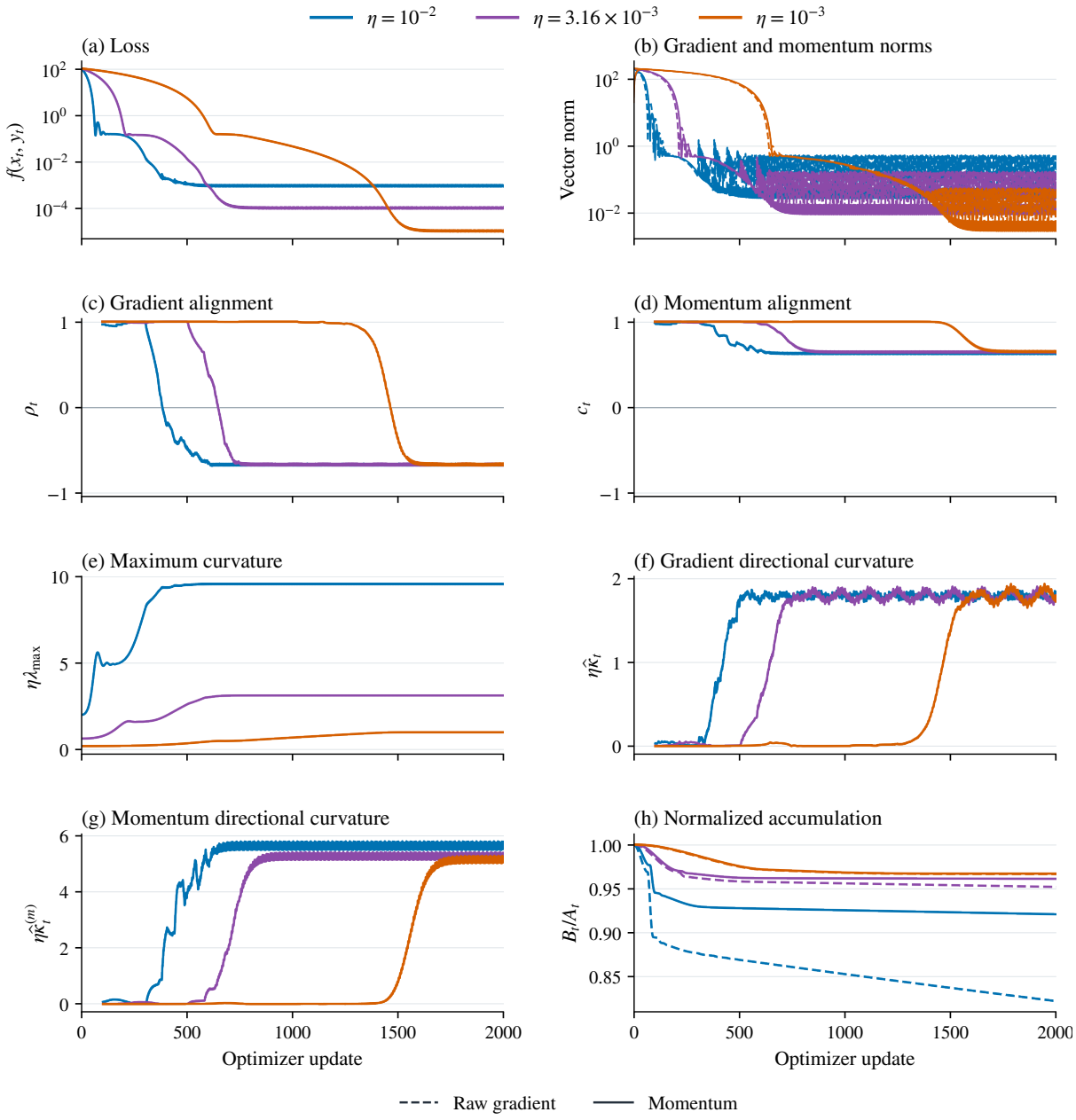


Figure 4: Rosenbrock diagnostics for the three learning rates. (a) Objective; (b) raw-gradient and stored-momentum norms (log Y); (c,d) adjacent cosines; (e) $\eta\lambda_{\max}$; (f,g) gradient- and momentum-derived curvature estimates; (h) full-prefix B/A, with raw gradients dashed. Panels (c,d,f,g) use complete 100-pair trailing means, first defined at update 101; other curves are unsmoothed. All 2,000 updates are retained.

4.1 Gradient-estimation noise

We repeat the same setup with $g_t = \nabla f(\theta_{t-1}) + \xi_t$, where $\xi_t \sim \mathcal{N}(0, \sigma^2 I_2)$ and $\sigma \in \{0, 0.01, 0.1, 1, 10\}$. One seed (1234) supplies shared standard-normal draws across noise levels and learning rates; $\sigma = 0$ is the deterministic control. In this paired realization, stronger noise raises the objective and reduces raw-gradient B/A, while momentum remains positively aligned on average (Fig. 5).

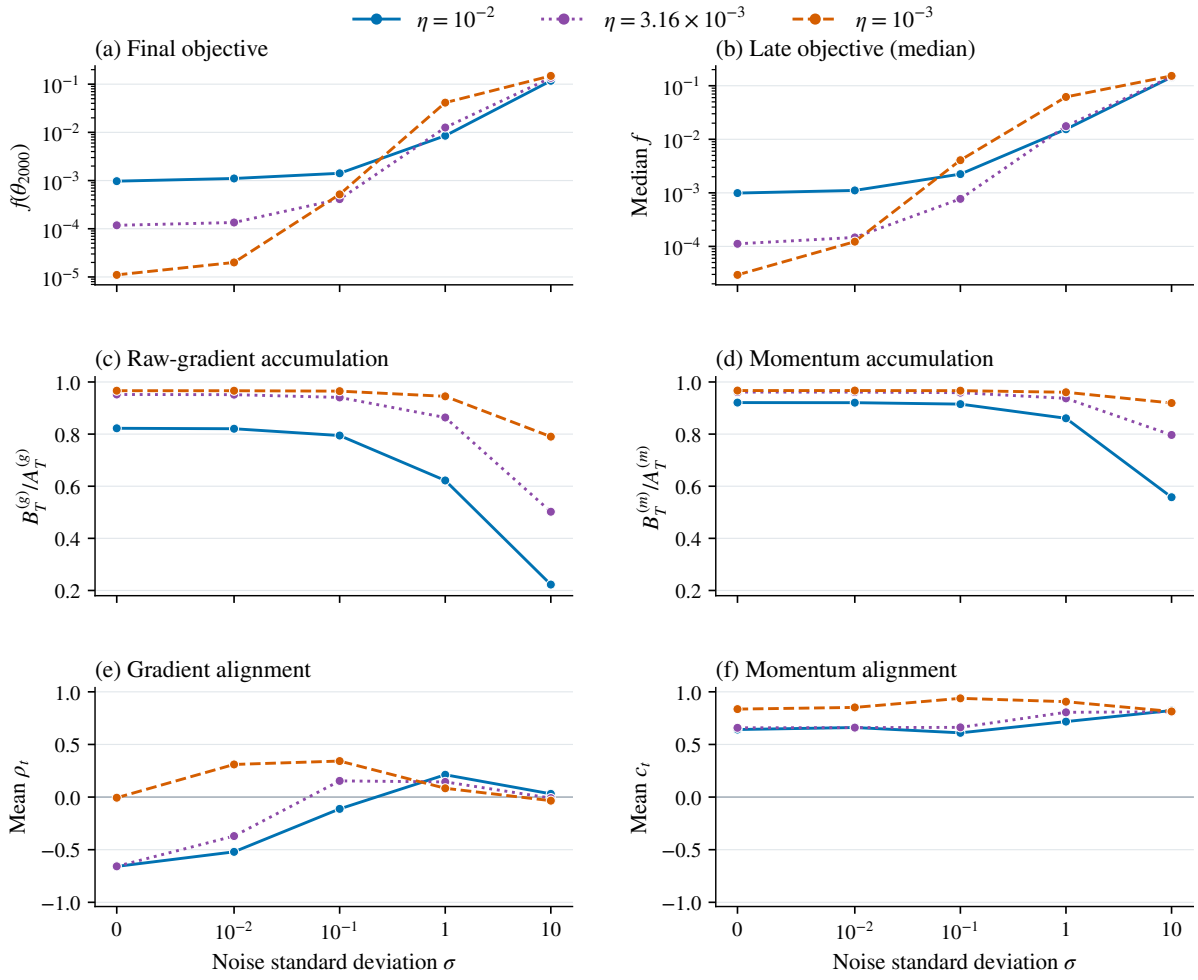


Figure 5: Gaussian gradient-noise sweep. (a) Final objective; (b) median objective over the last 1,000 updates; (c,d) full-prefix B/A for noisy gradients and stored momentum; (e,f) mean unsmoothed adjacent cosine over updates 1,001–2,000. Each marker is one trajectory; lines connect sampled noise levels. Objective uses log Y; the σ axis is symmetric log, linear through 0.01. One paired seed; no uncertainty bands.

References

[1] J. M. Cohen, S. Kaur, Y. Li, J. Z. Kolter, and A. Talwalkar. Gradient descent on neural networks typically occurs at the edge of stability. *International Conference on Learning Representations*, 2021. `https://arxiv.org/abs/2103.00065`.

[2] J. M. Cohen and collaborators. Released implementation of the edge-of-stability experiments, `github` branch. `https://github.com/locuslab/edge-of-stability/tree/github`.